INCH-POUND

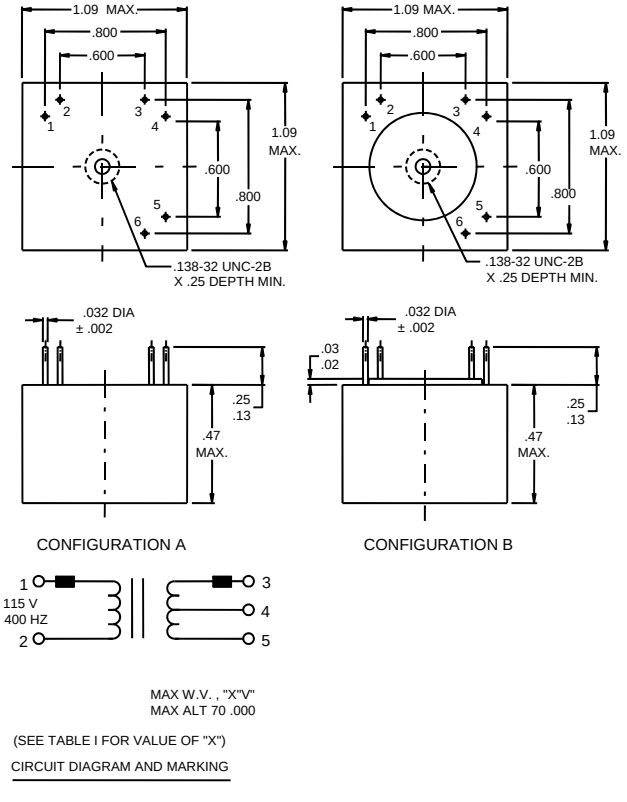
MIL-PRF-27/37D
5 December 2013
SUPERSEDING
MIL-PRF-27/37C
25 September 2009

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 1.5 VOLT AMPERES, 400 HZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification sheet and the latest issue of MIL-PRF-27.



Inches	mm
.002	0.05
.02	0.5
.03	0.8
.032	0.81
.13	3.3
.25	6.4
.47	11.9
.600	15.24
.75	19.1
.800	20.3
1.09	27.2

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.005 (0.13 mm).
4. Marking shall be on the side of the case.
5. A white dot shall be marked on the side opposite the terminals, over terminal 1, in addition to the terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: .85 ounce, maximum.

Duty cycle: Continuous.

Case: Plastic encapsulated.

Material: Plastic.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55˚C to +130˚C.

Terminals: Printed circuit pins - tinned brass.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts rms, 400 Hz across (1-2): Current in (1-2) shall not exceed 1.5 mA rms.

Rated load: With 115 volts rms, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 70 percent minimum at rated load.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{\text{Voltage (no Load) - Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$

shall not exceed 23 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 17°C, maximum.

Vibration: MIL-STD-202, method 204, test condition D.

Marking location: See figure 1.

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VERIFICATION

Extent of qualification

Qualification testing and approval to M27/37-44 A or B, M27/42-44 A or B, and M27/46-44 A or B shall be sufficient to grant qualification approval to MIL-PRF-27/37 through MIL-PRF-27/47 inclusive, all parts.

Qualification testing and approval to M27/37-44 shall be sufficient to grant qualification approval to M27/37-01 through M27/37-44.

Part or Identification Number (PIN): M27/37-01 A or B.

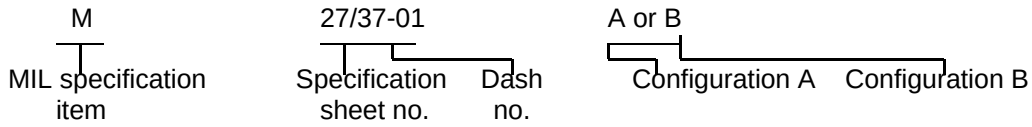

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TABLE I. Electrical ratings.

Dash No.	Type Designation	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms) (MAX)
01 A or B	400	6	0.250	325
02 A or B	401	6.3	0.238	325
03 A or B	402	7	0.214	325
04 A or B	403	7.5	0.200	325
05 A or B	404	8	0.188	325
06 A or B	405	9	0.167	325
07 A or B	406	10	0.150	325
08 A or B	407	11	0.136	325
09 A or B	408	12	0.125	325
10 A or B	409	12.6	0.119	325
11 A or B	410	13	0.115	325
12 A or B	411	15	0.100	325
13 A or B	412	16	0.094	325
14 A or B	413	18	0.083	325
15 A or B	414	20	0.075	325
16 A or B	415	22	0.068	325
17 A or B	416	24	0.063	325
18 A or B	417	26	0.058	325
19 A or B	418	28	0.053	325
20 A or B	419	30	0.050	325
21 A or B	420	33	0.045	325
22 A or B	421	36	0.042	325
23 A or B	422	39	0.038	325
24 A or B	423	43	0.035	325
25 A or B	424	47	0.032	325
26 A or B	425	51	0.029	325
27 A or B	426	56	0.027	325
28 A or B	427	62	0.024	325
29 A or B	428	68	0.022	325
30 A or B	429	75	0.020	325
31 A or B	430	82	0.018	325
32 A or B	431	91	0.016	325
33 A or B	432	100	0.015	325
34 A or B	433	110	0.014	325
35 A or B	434	115	0.013	325
36 A or B	435	120	0.0125	325
37 A or B	436	130	0.0115	325
38 A or B	437	150	0.010	325
39 A or B	438	180	0.0085	325
40 A or B	439	200	0.0075	440
41 A or B	440	220	0.007	440
42 A or B	441	240	0.006	440
43 A or B	442	270	0.0055	580
44 A or B	443	300	0.005	580

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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-PRF-27/38	MIL-PRF-27/39	MIL-PRF-27/40	MIL-PRF-27/41
MIL-PRF-27/42	MIL-PRF-27/43	MIL-PRF-27/44	MIL-PRF-27/45
MIL-PRF-27/46	MIL-PRF-27/47	MIL-STD-202.	

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2013-044)

Review activities:
Army - CR4, MI
Navy - AS, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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